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GATE 2022

ELECTRONICS ENGINEERING

Exam held on
06/02/2022

Questions & Solutions

Corporate Office : 44-A/1, Kalu Sarai, New Delhi - 110016

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SECTION - A

GENERAL APTITUDE

Q.1 Mr. X speaks _____ Japanese _____ Chinese.

- (a) neither / or (b) either / nor
(c) neither / nor (d) also / but

Ans. (c)

End of Solution

Q.2 A sum of money is to be distributed among P , Q , R , and S in the proportion $5 : 2 : 4 : 3$, respectively.

If R gets Rs. 1000 more than S , what is the share of Q (in Rs.)?

- (a) 500 (b) 1000
(c) 1500 (d) 2000

Ans. (d)

$$P = 5x, Q = 2x, R = 4x, S = 3x$$

Given: R gets Rs. 1000 more than S

$$4x - 3x = 1000$$

$$x = 1000$$

$$\text{Share of } Q = 2x = 2 \times 1000 = \text{Rs. } 2000$$

End of Solution

Q.3 A trapezium has vertices marked as P , Q , R and S (in that order anticlockwise). The side PQ is parallel to side SR .

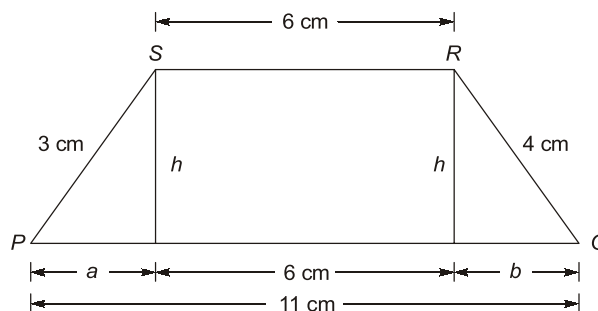
Further, it is given that, $PQ = 11$ cm, $QR = 4$ cm, $RS = 6$ cm and $SP = 3$ cm.

What is the shortest distance between PQ and SR (in cm)?

- (a) 1.80 (b) 2.40
(c) 4.20 (d) 5.76

Ans. (b)

Given :



Trapezium $PQRS$

h is distance between parallel sides,

$$a + b = 11 - 6 = 5$$



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From its diagram,

$$h^2 + b^2 = 16 \quad \dots(i)$$

$$h^2 + a^2 = 9 \quad \dots(ii)$$

$$b^2 - a^2 = 7$$

$$(b + a)(b - a) = 7$$

$$5(b - a) = 7$$

$$(b - a) = \frac{7}{5}$$

$$(b + a) = 5 \quad \dots(iii)$$

$$2b = \frac{7}{5} + 5 = \frac{32}{5}$$

$$b = \frac{16}{5}$$

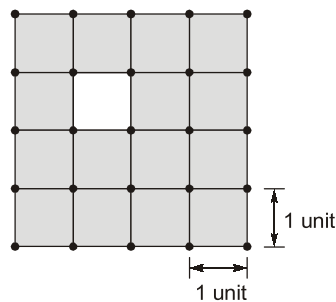
From equation (i),

$$h^2 = 16 - \left(\frac{16}{5}\right)^2 = 16 - \frac{256}{25} = \frac{400 - 256}{25} = \frac{144}{25}$$

$$h = \sqrt{\frac{144}{25}} = \frac{12}{5} = 2.4$$

End of Solution

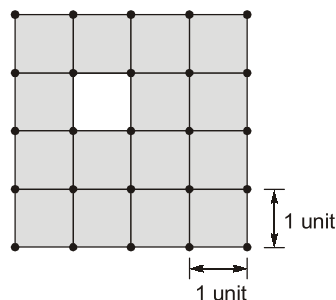
- Q.4** The figure shows a grid formed by a collection of unit squares. The unshaded unit square in the grid represents a hole.



What is the maximum number of squares without a “hole in the interior” that can be formed within the 4×4 grid using the unit squares as building blocks?

- (a) 15 (b) 20
(c) 21 (d) 26

Ans. (b)



Number of squares without hole

$$4 \times 4 - 1 = 16 - 1 = 15$$

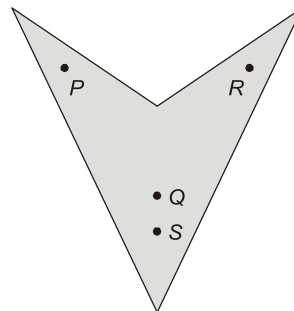
These are (1×1) squares

$$\text{Now } 2 \times 2 \text{ squares} = 5$$

$$\text{Total number of squares} = 15 + 5 = 20$$

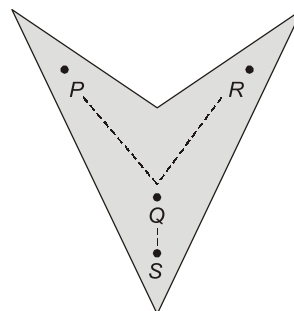
End of Solution

- Q.5** An art gallery engages a security guard to ensure that the items displayed are protected. The diagram below represents the plan of the gallery where the boundary walls are opaque. The location the security guard posted is identified such that all the inner space (shaded region in the plan) of the gallery is within the line of sight of the security guard. If the security guard does not move around the posted location and has a 360° view, which one of the following correctly represents the set of ALL possible locations among the locations P, Q, R and S, where the security guard can be posted to watch over the entire inner space of the gallery.



- (a) P and Q (b) Q
(c) Q and S (d) R and S

Ans. (c)



Security guard posting at Q and S can watch entire space of gallery.

End of Solution

Q.6 Mosquitoes pose a threat to human health. Controlling mosquitoes using chemicals may have undesired consequences. In Florida, authorities have used genetically modified mosquitoes to control the overall mosquito population. It remains to be seen if this novel approach has unforeseen consequences.

Which one of the following is the correct logical inference based on the information in the above passage?

- (a) Using chemicals to kill mosquitoes is better than using genetically modified mosquitoes because genetic engineering is dangerous.
- (b) Using genetically modified mosquitoes is better than using chemicals to kill mosquitoes because they do not have any side effects.
- (c) Both using genetically modified mosquitoes and chemicals have undesired consequences and can be dangerous.
- (d) Using chemicals to kill mosquitoes may have undesired consequences but it is not clear if using genetically modified mosquitoes has any negative consequence.

Ans. (d)

End of Solution

Q.7 Consider the following inequalities.

- (i) $2x - 1 > 7$
- (ii) $2x - 9 < 1$

Which one of the following expressions below satisfies the above two inequalities?

- (a) $x \leq -4$
- (b) $-4 < x \leq 4$
- (c) $4 < x < 5$
- (d) $x \geq 5$

Ans. (c)

$$\begin{array}{l|l} 2x - 1 > 7 & 2x - 9 < 1 \\ 2x > 8 & 2x < 10 \\ x > 4 & x < 5 \end{array}$$

$$\therefore 4 < x < 5.$$

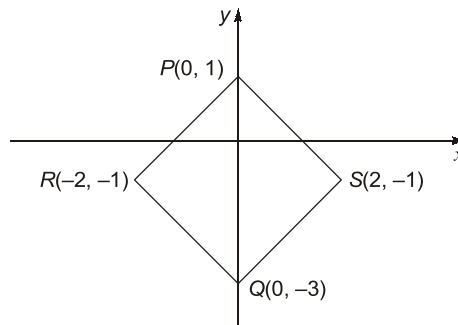
End of Solution

Q.8 Four points $P(0, 1)$, $Q(0, -3)$, $R(-2, -1)$, and $S(2, -1)$ represent the vertices of a quadrilateral. What is the area enclosed by the quadrilateral?

- (a) 4
- (b) $4\sqrt{2}$
- (c) 8
- (d) $8\sqrt{2}$

Ans. (c)

According to data quadrilateral $PSQR$



$$PS = \sqrt{(2-0)^2 + (-1-1)^2} = \sqrt{8}$$

$$QS = \sqrt{(2-0)^2 + (-1+3)^2} = \sqrt{8}$$

$$QR = \sqrt{(0+2)^2 + (-3+1)^2} = \sqrt{8}$$

$$PR = \sqrt{(-2-0)^2 + (1+1)^2} = \sqrt{8}$$

$\therefore$ All sides are equal

Diagonals,

$$PQ = \sqrt{(0-0)^2 + (1+3)^2} = 4$$

$$RS = \sqrt{(2+2)^2 + (-1+1)^2} = 4$$

Diagonals are equal

$\therefore$ It is a square

$$\text{Area of square} = (\sqrt{8}) \times (\sqrt{8}) = 8 \text{ square units}$$

End of Solution

Q.9 In a class of five students P, Q, R, S and T, only one student is known to have copied in the exam. The disciplinary committee has investigated the situation and recorded the statements from the students as given below.

Statement of P: R has copied in the exam.

Statement of Q: S has copied in the exam.

Statement of R: P did not copy in the exam.

Statement of S: Only one of us is telling the truth.

Statement of T: R is telling the truth.

The investigating team had authentic information that S never lies.

Based on the information given above, the person who has copied in the exam is

(a) R

(b) P

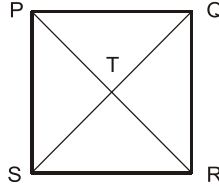
(c) Q

(d) T

Ans. (b)

End of Solution

- Q.10** Consider the following square with the four corners and the center marked as P, Q, R, S and T respectively.



Let X, Y and Z represent the following operations:

X: rotation of the square by 180 degree with respect to the S-Q axis.

Y: rotation of the square by 180 degree with respect to the P-R axis.

Z: rotation of the square by 90 degree clockwise with respect to the axis perpendicular, going into the screen and passing through the point T.

Consider the following three distinct sequences of operation (which are applied in the left to right order).

(1) XYZ

(2) XY

(3) ZZZ

Which one of the following statements is correct as per the information provided above?

- (a) The sequence of operations (1) and (2) are equivalent
- (b) The sequence of operations (1) and (3) are equivalent
- (c) The sequence of operations (2) and (3) are equivalent
- (d) The sequence of operations (1), (2) and (3) are equivalent

Ans. (b)

End of Solution



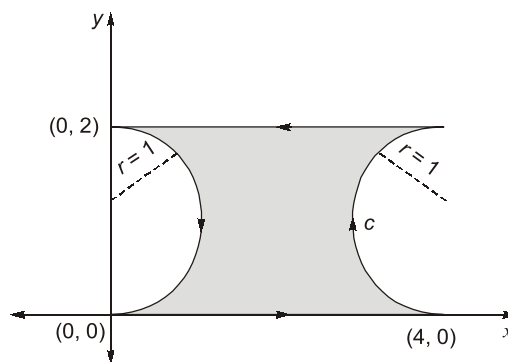
SECTION - B

TECHNICAL

- Q.11** Consider the two-dimensional vector field $\vec{F}(x, y) = x\vec{i} + y\vec{j}$, where $\vec{i}$ and $\vec{j}$ denote the unit vectors along the x -axis and the y -axis, respectively. A contour C in the x - y plane, as shown in the figure, is composed of two horizontal lines connected at the two ends by two semicircular arcs of unit radius. The contour is traversed in the counter-clockwise sense. The value of the closed path integral

$$\oint_C \vec{F}(x, y) \cdot (dx\vec{i} + dy\vec{j})$$

is _____.



- (a) 0
(b) 1
(c) $8 + 2\pi$
(d) -1

Ans. (a)

$$\oint_C \vec{F}(x, y) \cdot (dx\vec{i} + dy\vec{j})$$

$$= \oint_C (x\vec{i} + y\vec{j}) \cdot (dx\vec{i} + dy\vec{j}) = \oint_C xdx + ydy$$

By Green's theorem

$$= \iint_R (0 - 0) dx dy = 0$$

End of Solution

- Q.12** Consider a system of linear equations $Ax = b$, where

$$A = \begin{bmatrix} 1 & -\sqrt{2} & 3 \\ -1 & \sqrt{2} & -3 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

This system of equations admits _____.

- (a) a unique solution for x
(b) infinitely many solutions for x
(c) no solutions for x
(d) exactly two solutions for x

Ans. (c)

$$Ax = B$$

$$x - \sqrt{2}y + 3z = 1 \quad (\text{Inconsistent})$$

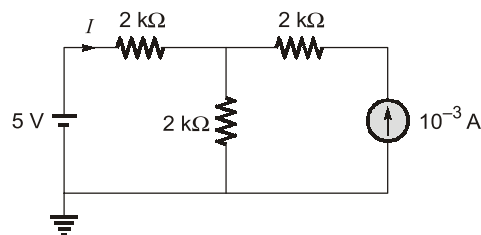
$$-x + \sqrt{2}y - 3z = 3 \Rightarrow x - \sqrt{2}y + 3z = -3 \neq 1 \quad (\text{Inconsistent})$$

∴ Both lines are parallel to each other.

∴ It has no solution.

End of Solution

Q.13 The current I in the circuit shown is _____.



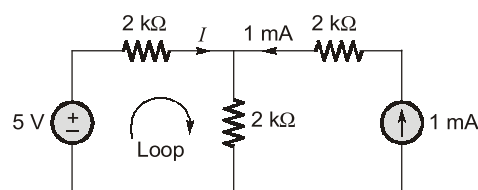
(a) $1.25 \times 10^{-3} \text{ A}$

(b) $0.75 \times 10^{-3} \text{ A}$

(c) $-0.5 \times 10^{-3} \text{ A}$

(d) $1.16 \times 10^{-3} \text{ A}$

Ans. (b)



Applying KVL in the loop,

$$5 = 2kI + 2k(I + 10^{-3})$$

$$5 = I(4k) + 2 \times 10^3 \times 1 \times 10^{-3}$$

$$I(4k) = 5 - 2$$

$$I = \frac{3}{4k}$$

$$I = 0.75 \times 10^{-3} \text{ A}$$

$$I = 0.75 \text{ mA}$$

End of Solution



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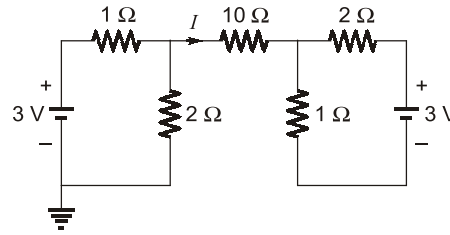
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- Q.14** Consider the circuit shown in the figure. The current I flowing through the $10\ \Omega$ resistor is _____.



- (a) 1 A
(b) 0 A
(c) 0.1 A
(d) -0.1 A

Ans. (b)

As there is no return path for current,
 $\therefore I = 0$

End of Solution

- Q.15** The Fourier transform $X(j\omega)$ of the signal

$$x(t) = \frac{t}{(1+t^2)^2}$$

is _____.

- (a) $\frac{\pi}{2j} \omega e^{-|\omega|}$
(b) $\frac{\pi}{2} \omega e^{-|\omega|}$
(c) $\frac{\pi}{2j} e^{-|\omega|}$
(d) $\frac{\pi}{2} e^{-|\omega|}$

Ans. (a)

Consider, $x(t) = e^{-|t|}$
by taking Fourier transform,

$$e^{-|t|} \xleftrightarrow{F.T} \frac{2}{1+\omega^2}$$

Apply differentiation in frequency,

$$tx(t) \xleftrightarrow{F.T} j \frac{d}{d\omega} X(\omega)$$

$$te^{-|t|} \xleftrightarrow{F.T} j \left[\frac{d}{d\omega} \left(\frac{2}{1+\omega^2} \right) \right]$$

$$te^{-|t|} \xleftrightarrow{F.T} j \left[\frac{[0 - 2(2\omega)]}{(1+\omega^2)^2} \right]$$

$$te^{-|t|} \xleftrightarrow{F.T} \frac{-4j\omega}{(1+\omega^2)^2}$$

Apply duality property,

$$\begin{aligned}\frac{-4jt}{(1+t^2)^2} &\xrightarrow{F.T} 2\pi(-\omega)e^{-|\omega|} \\ \frac{t}{(1+t^2)^2} &\xrightarrow{F.T} \frac{-2\pi\omega e^{-|\omega|}}{-4j} \\ \frac{t}{(1+t^2)^2} &\xrightarrow{F.T} \frac{\pi}{2j}\omega e^{-|\omega|}\end{aligned}$$

End of Solution

- Q.16** Consider a long rectangular bar of direct bandgap p-type semiconductor. The equilibrium hole density is 10^{17} cm^{-3} and the intrinsic carrier concentration is 10^{10} cm^{-3} . Electron and hole diffusion lengths are $2 \mu\text{m}$ and $1 \mu\text{m}$, respectively.

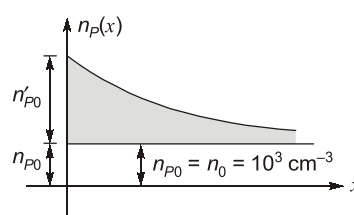
The left side of the bar ($x = 0$) is uniformly illuminated with a laser having photon energy greater than the bandgap of the semiconductor. Excess electron-hole pairs are generated ONLY at $x = 0$ because of the laser. The steady state electron density at $x = 0$ is 10^{14} cm^{-3} due to laser illumination. Under these conditions and ignoring electric field, the closest approximation (among the given options) of the steady state electron density at $x = 2 \mu\text{m}$, is _____.

- (a) $0.37 \times 10^{14} \text{ cm}^{-3}$ (b) $0.63 \times 10^{13} \text{ cm}^{-3}$
(c) $3.7 \times 10^{14} \text{ cm}^{-3}$ (d) 10^3 cm^{-3}

Ans. (a)

Given,

$$\begin{aligned}P &= 10^{17} \text{ cm}^{-3} = P_0 \\ n_0 &= \frac{n_i^2}{P_0} = \frac{10^{20}}{10^{17}} = 10^3 \text{ cm}^{-3} \\ L_n &= 2 \mu\text{m} \\ n'_{P0} &= 10^{14} \text{ cm}^{-3}\end{aligned}$$



Excess electron concentration at any distance x is

$$\begin{aligned}\delta n_p(x) &= n'_{P0} e^{-x/L_n} \\ &= 10^{14} e^{-2/2} \\ &= 10^{14} e^{-1} \\ &= 3.67 \times 10^{13} \text{ cm}^{-3} \\ &= 0.367 \times 10^{14} \text{ cm}^{-3}\end{aligned}$$

End of Solution

- Q.17** In a non-degenerate bulk semiconductor with electron density $n = 10^{16} \text{ cm}^{-3}$, the value of $E_C - E_{Fn} = 200 \text{ meV}$, where E_C and E_{Fn} denote the bottom of the conduction band energy and electron Fermi level energy, respectively. Assume thermal voltage as 26 meV and the intrinsic carrier concentration is 10^{10} cm^{-3} . For $n = 0.5 \times 10^{16} \text{ cm}^{-3}$, the closest approximation of the value of $(E_C - E_{Fn})$, among the given options, is _____.
- (a) 226 meV (b) 174 meV
(c) 218 meV (d) 182 meV

Ans. (c)

$$\text{Given, } (E_C - E_{Fn})_1 = kT \ln \left(\frac{N_C}{N_{D1}} \right) = 200 \text{ meV} \quad \dots(1)$$

$$(E_C - E_{Fn})_2 = kT \ln \left(\frac{N_C}{N_{D2}} \right) \quad \dots(2)$$

$$N_{D1} = 10^{16} \text{ cm}^{-3}$$

$$N_{D2} = 0.5 \times 10^{16} \text{ cm}^{-3}$$

$$(E_C - E_{Fn})_1 - (E_C - E_{Fn})_2 = kT \ln \left(\frac{N_C}{N_{D1}} \right) - kT \ln \left(\frac{N_C}{N_{D2}} \right)$$

$$200 \text{ meV} - (E_C - E_{Fn})_2 = kT \ln \left(\frac{N_{D2}}{N_{D1}} \right)$$

$$= 0.026 \ln \left(\frac{10^{16} \times 0.5}{10^{16}} \right)$$

$$= 0.026 \ln(0.5)$$

$$= -0.01802 \text{ volt}$$

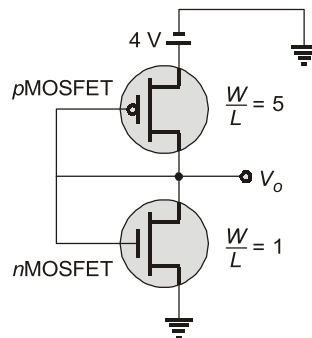
$$= -18.02 \text{ meV}$$

$$(E_C - E_{Fn})_2 = 200 \text{ meV} + 18.02 \text{ meV}$$

$$= 218.02 \text{ meV}$$

End of Solution

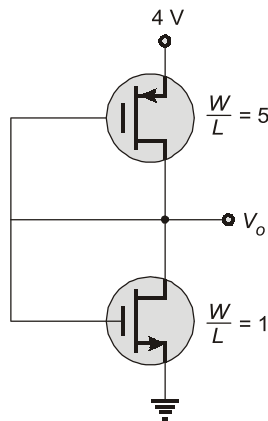
- Q.18** Consider the CMOS circuit shown in the figure (substrates are connected to their respective sources). The gate width (W) to gate length (L) ratios $\left(\frac{W}{L} \right)$ of the transistors are as shown. Both the transistors have the same gate oxide capacitance per unit area. For the pMOSFET, the threshold voltage is -1 V and the mobility of holes is $40 \frac{\text{cm}^2}{\text{V.s}}$. For the nMOSFET, the threshold voltage is 1 V and the mobility of electrons is $300 \frac{\text{cm}^2}{\text{V.s}}$. The steady state output voltage V_o is _____.



- (a) equal to 0 V
(b) more than 2 V
(c) less than 2 V
(d) equal to 2 V

Ans. (c)

Both MOSFETs are in saturation because drain is shorted to Gate.



$$I_{DSN} = I_{SDP}$$

$$\frac{\mu_n C_{ox}}{2} \times \left(\frac{W}{L}\right)_N (V_{GSN} - V_{TN})^2 = \frac{\mu_p C_{ox}}{2} \left(\frac{W}{L}\right)_P (V_{SGP} - |V_{TP}|)^2$$

$$300 \times 1 (V_o - 1)^2 = 40 \times 5 (4 - V_o - 1)^2$$

$$3(V_o^2 + 1 - 2V_o) = 2(9 + V_o^2 - 6V_o)$$

$$\Rightarrow V_o^2 + 6V_o - 15 = 0$$

$$V_o = \frac{-6 \pm \sqrt{36 + 4 \times 15}}{2} = \frac{-6 \pm \sqrt{96}}{2}$$

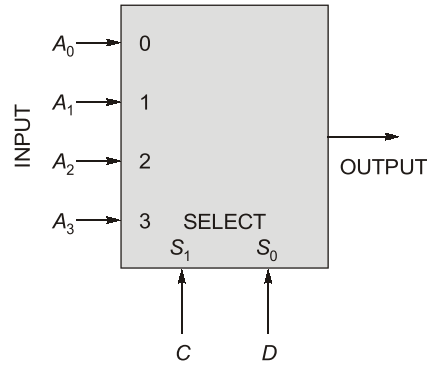
$$V_o = 1.898 \text{ V}, -7.89 \text{ V}$$

V_o cannot be negative because V_o should lie between 0 and 4 V.

$$\therefore V_o = 1.898 \text{ V}$$

End of Solution

- Q.19** Consider the 2-bit multiplexer (MUX) shown in the figure. For OUTPUT to be the XOR of C and D, the values for A_0 , A_1 , A_2 and A_3 are _____.



- (a) $A_0 = 0, A_1 = 0, A_2 = 1, A_3 = 1$ (b) $A_0 = 1, A_1 = 0, A_2 = 1, A_3 = 0$
(c) $A_0 = 0, A_1 = 1, A_2 = 1, A_3 = 0$ (d) $A_0 = 1, A_1 = 1, A_2 = 0, A_3 = 0$

Ans. (c)

The output of MUX, F is

$$F = \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3$$

$$F = \bar{C} \bar{D} A_0 + \bar{C} D A_1 + C \bar{D} A_2 + C D A_3$$

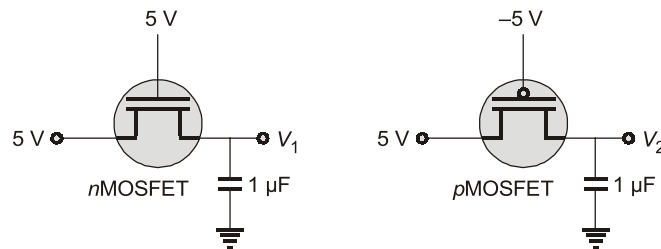
But output,

$$F = C \oplus D = \bar{C} D + C \bar{D}$$

$\therefore$ Inputs of MUX are, $A_0 = 0, A_1 = 1, A_2 = 1, A_3 = 0$

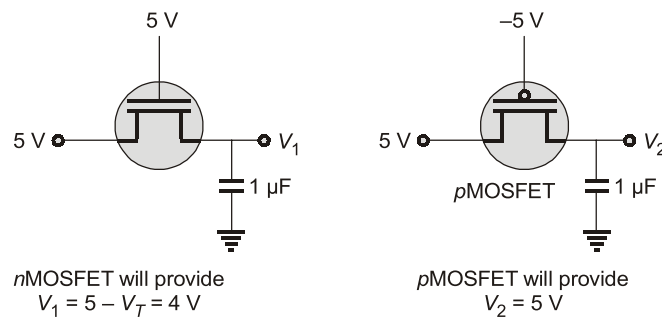
End of Solution

- Q.20** The ideal long channel n MOSFET and p MOSFET devices shown in the circuits have threshold voltages of 1 V and -1 V, respectively. The MOSFET substrates are connected to their respective sources. Ignore leakage currents and assume that the capacitors are initially discharged. For the applied voltages as shown, the steady state voltages are _____.



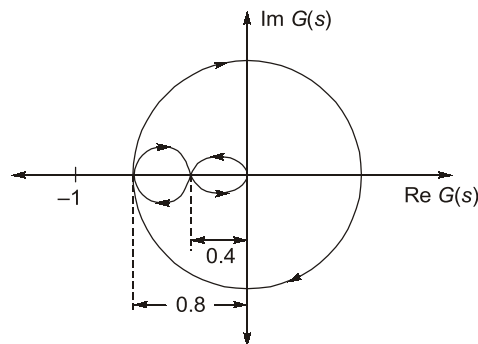
- (a) $V_1 = 5 \text{ V}, V_2 = 5 \text{ V}$ (b) $V_1 = 5 \text{ V}, V_2 = 4 \text{ V}$
(c) $V_1 = 4 \text{ V}, V_2 = 5 \text{ V}$ (d) $V_1 = 4 \text{ V}, V_2 = -5 \text{ V}$

Ans. (c)



End of Solution

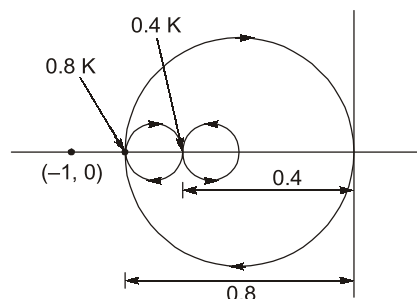
Q.21 Consider a closed-loop control system with unity negative feedback and $KG(s)$ in the forward path, where the gain $K = 2$. The complete Nyquist plot of the transfer function $G(s)$ is shown in the figure. Note that the Nyquist contour has been chosen to have the clockwise sense. Assume $G(s)$ has no poles on the closed right-half of the complex plane. The number of poles of the closed-loop transfer function in the closed right-half of the complex plane is _____.



- (a) 0
(b) 1
(c) 2
(d) 3

Ans. (c)

For, $K = 1$



Ans. (a)

There are 5 root locus branches from the same point, so there are 5 real multiple poles.

So, correct option is $\frac{1}{(s+1)^5}$.

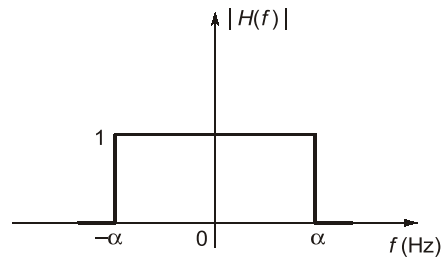
End of Solution

Q.23 The frequency response $H(f)$ of a linear time-invariant system has magnitude as shown in the figure.

Statement I: The system is necessarily a pure delay system for inputs which are bandlimited to $-\alpha \leq f \leq \alpha$.

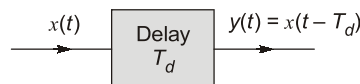
Statement II: For any wide-sense stationary input process with power spectral density $S_X(f)$, the output power spectral density $S_Y(f)$ obeys $S_Y(f) = S_X(f)$ for $-\alpha \leq f \leq \alpha$.

Which one of the following combinations is true?



- (a) Statement I is correct, Statement II is correct
- (b) Statement I is correct, Statement II is incorrect
- (c) Statement I is incorrect, Statement II is correct
- (d) Statement I is incorrect, Statement II is incorrect

Ans. (a)



$$Y(f) = X(f) \cdot e^{-j2\pi f T_d}$$

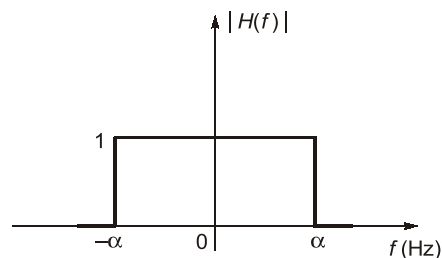
$$H(f) = \frac{Y(f)}{X(f)} = e^{-j2\pi f T_d}$$

$$|H(f)| = 1$$

$$\angle H(f) = -2\pi f T_d$$

Given that input is bandlimited to $-\alpha \leq f \leq \alpha$.

Magnitude response of the system given



Based on the magnitude response given ; statement (I) is correct.

Note : Phase response of the system is not given in the question without knowing phase response of the system we can't comment about exact nature of the system.

$$(\text{PSD})_{o/p} = (\text{PSD})_{i/p} \cdot |H(f)|^2$$

$$S_Y(f) = S_X(f) \quad ; \quad -\alpha \leq f \leq \alpha$$

Statement (II) is correct.

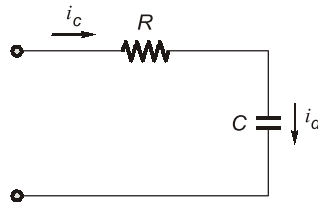
Hence option (a) is correct.

End of Solution

Q.24 In a circuit, there is a series connection of an ideal resistor and an ideal capacitor. The conduction current (in Amperes) through the resistor is $2\sin(t + \pi/2)$. The displacement current (in Amperes) through the capacitor is _____.

- (a) $2\sin(t)$ (b) $2\sin(t + \pi)$
(c) $2\sin(t + \pi/2)$ (d) 0

Ans. (c)



In series connection, current through each elements remain same. Hence, $i_c = i_d$.

So,

$$i_d = 2\sin\left(t + \frac{\pi}{2}\right)$$

End of Solution

Q.25 Consider the following partial differential equation (PDE)

$$a \frac{\partial^2 f(x, y)}{\partial x^2} + b \frac{\partial^2 f(x, y)}{\partial y^2} = f(x, y),$$

where a and b are distinct positive real numbers. Select the combination(s) of values of the real parameters ξ and η such that $f(x, y) = e^{(\xi x + \eta y)}$ is a solution of the given PDE.

- (a) $\xi = \frac{1}{\sqrt{2a}}, \eta = \frac{1}{\sqrt{2b}}$ (b) $\xi = \frac{1}{\sqrt{a}}, \eta = 0$
(c) $\xi = 0, \eta = 0$ (d) $\xi = \frac{1}{\sqrt{a}}, \eta = \frac{1}{\sqrt{b}}$

Ans. (a, b)

$$f(x, y) = e^{\xi x + \eta y}$$

Differentiating $f(x, y)$ two times w.r.t. x .

$$\frac{\partial^2 f(x, y)}{\partial x^2} = \xi^2 e^{\xi x + \eta y}$$

Differentiating $f(x, y)$ two times w.r.t. y ,

$$\frac{\partial^2 f(x, y)}{\partial y^2} = \eta^2 e^{\xi + \eta y}$$

$$\frac{a \partial^2 f(x, y)}{\partial x^2} + \frac{b \partial^2 f(x, y)}{\partial y^2} = (a\xi^2 + b\eta^2) f(x, y)$$

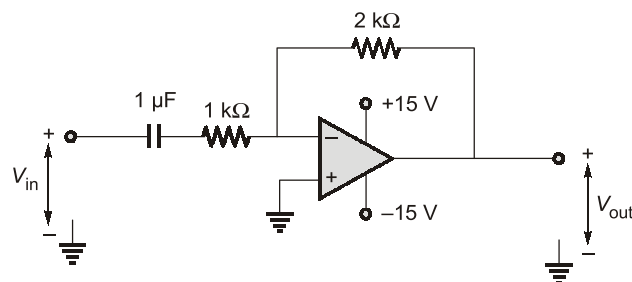
$$a\xi^2 + b\eta^2 = 1$$

For $\xi = \frac{1}{\sqrt{a}}, \eta = 0$

$$a \times \left(\frac{1}{\sqrt{a}}\right)^2 + b \times 0 = \frac{a}{a} = 1$$

End of Solution

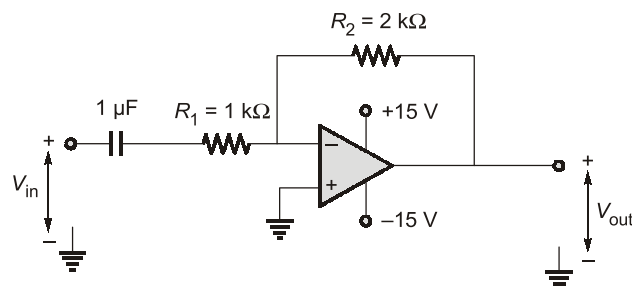
- Q.26** An ideal OPAMP circuit with a sinusoidal input is shown in the figure. The 3 dB frequency is the frequency at which the magnitude of the voltage gain decreases by 3 dB from the maximum value. Which of the options is/are correct?



- (a) The circuit is a low pass filter. (b) The circuit is a high pass filter.
(c) The 3 dB frequency is 1000 rad/s. (d) The 3 dB frequency is $\frac{1000}{3}$ rad/s.

Ans. (b, c)

Given circuit is a high pass filter.



$$\omega_c = \frac{1}{R_1 C_1} = \frac{1}{10^3 \times 10^{-6}} = 1000 \text{ rad/sec}$$

Hence, options (b) and (c) is correct.

End of Solution



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Q.27 Select the Boolean function(s) equivalent to $x + yz$, where x , y , and z are Boolean variables, and $+$ denotes logical OR operation.

- (a) $x + z + xy$ (b) $(x + y)(x + z)$
(c) $x + xy + yz$ (d) $x + xz + xy$

Ans. (b, c)

Given : $f = x + yz$
(a) $x + xy + xz = x(1 + y) + xz = x + xz = x$
(b) $(x + y)(x + z) = x + yz$
(c) $x + xy + yz = x(1 + y) + yz = x + yz$
(d) $xy + yz = y(x + z)$
So, option (b, c) are correct.

End of Solution

Q.28 Select the correct statement(s) regarding CMOS implementation of NOT gates.

- (a) Noise Margin High (NM_H) is always equal to the Noise Margin Low (NM_L), irrespective of the sizing of transistors.
(b) Dynamic power consumption during switching is zero.
(c) For a logical high input under steady state, the nMOSFET is in the linear regime of operation.
(d) Mobility of electrons never influences the switching speed of the NOT gate.

Ans. (c)

(a) $NM_L = V_{IL} - V_{OL}$
 $NM_H = V_{OH} - V_{IH}$

when, $V_{TN} = |V_{TP}|$

and $K_n = K_p$

$V_{IT} = \frac{V_{DD}}{2}$ and $NM_L = NM_H$

when $\uparrow \frac{K_p}{K_n} > 1, V_{IL} \uparrow, V_{IH} \uparrow$

$NM_H \downarrow$ and $NM_L \uparrow$

When, $\downarrow \frac{K_p}{K_n} < 1, V_{IL} \downarrow, V_{IH} \uparrow$

$NM_H \uparrow$ and $NM_L \uparrow$

i.e. NM_L and NM_H depends on transistor sizing and they are equal for certain condition only.

- (b) Dynamic power consumption during switching is non-zero due to capacitive loading of next stage.

- (c) For $V_{DD} - |V_{TP}| \leq V_{in} \leq V_{DD} \Rightarrow$ PMOS $\rightarrow$ Cut-off
(logic high input) NMOS $\rightarrow$ Linear

- (d) Switching speed depends on charging and discharging of load capacitor for pull up and pull down of output voltage respectively.

Fast charging and discharging of capacitor depends on mobility of charge carrier.

$$\tau_{\text{charging}} = R_{\text{PMOS}} C_L \text{ (to bring o/p at logic high)}$$

$$\tau_{\text{discharging}} = R_{\text{NMOS}} C_L \text{ (to bring o/p at logic low)}$$

and also propagation delay,

$$\tau_p = \frac{\tau_{\text{PLH}} + \tau_{\text{PHL}}}{2}$$

where,

$$\tau_{\text{PLH}} = \frac{C_L V_{DD}}{\mu_p C_{ox} \left(\frac{W}{L}\right) (V_{GS} - V_{Tp})^2}$$

$$\tau_{\text{PHL}} = \frac{C_L V_{DD}}{\mu_p C_{ox} \left(\frac{W}{L}\right) (V_{GS} - V_{TN})^2}$$

Using average charge model.

Hence, (c) is only correct statement.

End of Solution

Q.29 Let $H(X)$ denote the entropy of a discrete random variable X taking K possible distinct real values. Which of the following statements is/are necessarily true?

- (a) $H(X) \leq \log_2 K$ bits (b) $H(X) \leq H(2X)$
(c) $H(X) \leq H(X^2)$ (d) $H(X) \leq H(2^X)$

Ans. (a, b, d)

Given that X is a discrete random variable taking K possible distinct real values. If 'K' symbols having equal probability then entropy will be maximum $H(X)_{\text{max}} = \log_2 K$. If symbols having different probabilities then $H(X) < \log_2 K$

So, that, $H(X) \leq \log_2 K$

Option (a) is correct.

Given option (b), $H(X) \leq H(2X)$

Let,

$X \in \{x_i\}$	-1	0	1
$P_X(x_i)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

$$\begin{aligned} H(X) &= \sum_i P_X(x_i) \log_2 \frac{1}{P_X(x_i)} \\ &= \frac{1}{4} \log_2 4 + \frac{1}{2} \log_2 2 + \frac{1}{4} \log_2 4 \end{aligned}$$

$$H(X) = 1.5 \frac{\text{bits}}{\text{symbol}}$$

Let,

$$Y = 2X$$

X	Y	P(Y)
-1	-2	1/4
0	0	1/2
1	2	1/4

$Y \in \{y_i\}$	-2	0	2
$P_Y(y_i)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

$$H(2X) = H(Y) = \sum_i P_Y(y_i) \log_2 \frac{1}{P_Y(y_i)} = 1.5 \frac{\text{bits}}{\text{symbol}}$$

For $Y = 2X$, distinct X values results in distinct 'Y' values so that $H(X) = H(Y)$.

So, option (b) is true i.e., $H(X) = H(2X)$

Given option (c), $H(X) \leq H(X^2)$;

Let $Y = X^2$,

X	Y	P(Y)
-1	1	1/4
0	0	1/2
1	1	1/4

$Y \in \{y_i\}$	0	1
$P(Y = y_i)$	$\frac{1}{2}$	$\frac{1}{2}$

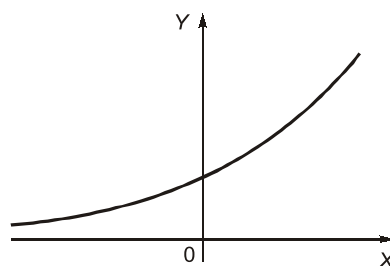
$$\begin{aligned} H(X^2) &= H(Y) = \sum_i P_Y(y_i) \log_2 \frac{1}{P_Y(y_i)} \\ &= \frac{1}{2} \log_2 2 + \frac{1}{2} \log_2 2 = 1 \frac{\text{bit}}{\text{symbol}} \end{aligned}$$

whereas, $H(X) = 1.5 \frac{\text{bits}}{\text{symbol}}$

Option (c) is incorrect.

Given option (d) $H(X) \leq H(2^X)$

Let $Y = 2^X$



Here distinct 'X' values results in distinct 'Y' values.

So that, $H(X) = H(Y)$

i.e., $H(X) = H(2^X)$

Option (d) is true.

End of Solution

Q.30 Consider the following wave equation,

$$\frac{\partial^2 f(x, t)}{\partial t^2} = 10000 \frac{\partial^2 f(x, t)}{\partial x^2}$$

Which of the given options is/are solution(s) to the given wave equation?

- (a) $f(x, t) = e^{-(x-100t)^2} + e^{-(x+100t)^2}$ (b) $f(x, t) = e^{-(x-100t)} + 0.5e^{-(x+1000t)}$
(c) $f(x, t) = e^{-(x-100t)} + \sin(x+100t)$ (d) $f(x, t) = e^{j100\pi(-100x+t)} + e^{j100\pi(100x+t)}$

Ans. (a, c)

Given wave equation,

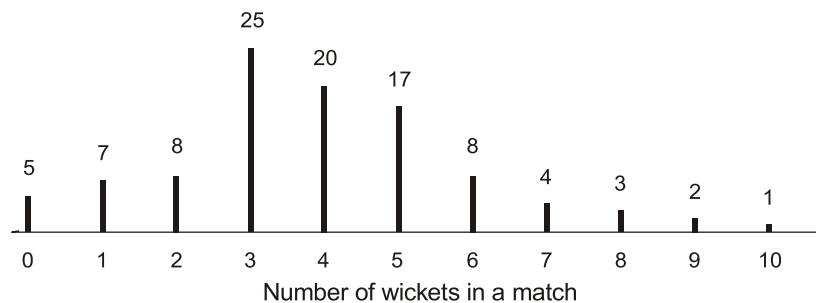
$$\frac{\partial^2 f(x, t)}{\partial t^2} = c^2 \frac{d^2 f}{dt^2} ; c = 100$$

Solution is given as $F = f(x \pm ct)$.

Hence, (a) and (c) satisfies the above solution.

End of Solution

Q.31 The bar graph shows the frequency of the number of wickets taken in a match by a bowler in her career. For example, in 17 of her matches, the bowler has taken 5 wickets each. The median number of wickets taken by the bowler in a match is _____ (rounded off to one decimal place).



Ans. (4)

$$\Sigma f = 5 + 7 + 8 + 25 + 20 + 17 + 8 + 4 + 3 + 2 + 1 = 100$$

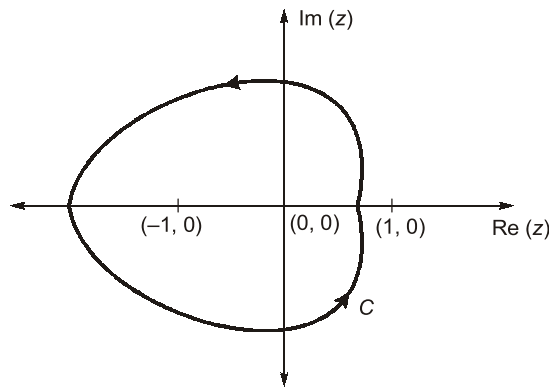
$$\text{Median} = \text{Average of } 50^{\text{th}} \text{ and } 51^{\text{st}} \text{ matches} = 4$$

End of Solution

Q.32 A simple closed path C in the complex plane is shown in the figure. If

$$\oint_C \frac{2^z}{z^2 - 1} dz = -i\pi A,$$

where $i = \sqrt{-1}$, then the value of A is _____ (rounded off to two decimal places).



Ans. (0.5)

Poles are given as,

$$z^2 - 1 = 0$$

$$z = \pm 1$$

$$\therefore \oint_C f(z) \cdot dz = 2\pi i [\text{Residue at } -1]$$

$$= 2\pi i \frac{2^{-1}}{(-1-1)} = \frac{-2\pi i}{2 \times 2}$$

$$= -\frac{\pi i}{2}$$

Comparing with $-i\pi A$

$$A = \frac{1}{2}$$

End of Solution

Q.33 Let $x_1(t) = e^{-t} u(t)$ and $x_2(t) = u(t) - u(t-2)$, where $u(\cdot)$ denotes the unit step function. If $y(t)$ denotes the convolution of $x_1(t)$ and $x_2(t)$, then $\lim_{t \rightarrow \infty} y(t) =$ _____ (rounded off to one decimal place).

Ans. (0)

Given that,

$$x_1(t) = e^{-2t} u(t)$$

$$x_2(t) = u(t) - u(t-2)$$

$$y(t) = x_1(t) * x_2(t)$$

By applying Laplace transform

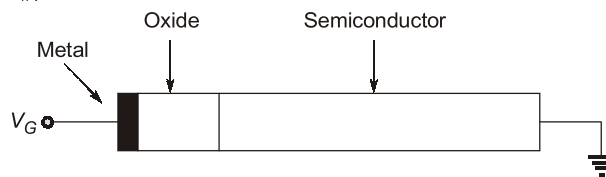
$$Y(s) = X_1(s) \cdot X_2(s) = \frac{1}{(s+1)} \frac{(1-e^{-2s})}{s}$$

By applying final value theorem,

$$y(t)|_{t=\infty} = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} \left(\frac{1 - e^{-2s}}{s+1} \right) = 0$$

End of Solution

- Q.34** An ideal MOS capacitor (p -type semiconductor) is shown in the figure. The MOS capacitor is under strong inversion with $V_G = 2$ V. The corresponding inversion charge density (Q_{in}) is $2.2 \mu\text{C}/\text{cm}^2$. Assume oxide capacitance per unit area as $C_{ox} = 1.7 \mu\text{F}/\text{cm}^2$. For $V_G = 4$ V, the value of Q_{in} is _____ $\mu\text{C}/\text{cm}^2$ (rounded off to one decimal place).



Ans. (5.6)

$$Q_{in} = C_{ox}(V_{GS} - V_T) \quad \text{or} \quad C_{ox}(V_G - V_T)$$

$$\frac{Q_{in}}{C_{ox}} = V_G - V_T$$

$$V_T = V_G - \frac{Q_{in}}{C_{ox}} = 2 - \frac{2.2}{1.7} = 2 - 1.294$$

$$V_T = 0.706 \text{ volt}$$

Now,

$$Q_{in} \text{ at } V_G = 4 \text{ V}$$

$$Q_{in} = C_{ox}(V_G - V_T) = 1.7 \times 10^{-6} \times (4 - 0.706)$$

$$Q_{in} = 5.5998 \mu\text{C}/\text{cm}^2 \approx 5.6 \mu\text{C}/\text{cm}^2$$

End of Solution

- Q.35** A symbol stream contains alternate QPSK and 16-QAM symbols. If symbols from this stream are transmitted at the rate of 1 mega-symbols per second, the raw (uncoded) data rate is _____ mega-bits per second (rounded off to one decimal place).

Ans. (3)

Given that symbol stream contains alternate QPSK and 16-QAM symbols.
Symbol rate given as 1 mega symbols per second.

$$\text{Bit rate} = \text{Symbol rate} \times \log_2 M$$

For QPSK, $\text{Bit rate} = 1 \text{M} \frac{\text{symbol}}{\text{sec}} \times \log_2 4 \frac{\text{bits}}{\text{symbol}}$

$$\text{Bit rate} = 2 \text{ Mbps}$$

For 16-QAM, $\text{Bit rate} = 1 \text{M} \frac{\text{symbol}}{\text{sec}} \times \log_2 16 \frac{\text{bits}}{\text{symbol}}$